

MVP Simulator: MDP Formulation and Initial Results

William Dennis

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1 MDP Formulation

The MVP simulator implements a finite-horizon MDP $(\mathcal{S}, \mathcal{A}, P, r, \gamma, T)$:

1.1 State Space

$$\mathcal{S} = \{\text{sedentary}, \text{active}\}$$

Two activity states with no additional time-of-day dimension. Time-dependent reward scaling is handled via `reward_multiplier_by_step` (see §1.4 below).

1.2 Action Space

$$\mathcal{A} = \{\text{nudge}, \text{idle}\}$$

A binary intervention: send a motivational message (`nudge`) or do not (`idle`).

1.3 Transition Function

$P(s' \mid s, a)$ is defined by a configurable matrix. The probabilities encode a Reactance effect: reminders help sedentary users but can backfire for users who are already active.

Current state s	Action a	$P(\text{active})$	$P(\text{sedentary})$
sedentary	nudge	0.3	0.7
sedentary	idle	0.1	0.9
active	nudge	0.5	0.5
active	idle	0.6	0.4

Table 1: Transition probabilities. Sending a reminder (`nudge`) increases the probability of becoming active for sedentary users ($0.3 > 0.1$) but decreases it for active users ($0.5 < 0.6$), consistent with Reactance theory.

Optional `reward_multiplier_by_step`: a per-step scaling factor that can zero out rewards at certain daily steps (soft mask). When absent, all multipliers default to 1.0. The per-step reward is computed as $\text{base reward} \times \text{multiplier}[t]$ where t is the step-of-day index.

1.4 Reward

$$r(s, a, t) = \text{per_step_reward}[t][s']$$

where s' is the next state after taking action a in state s , and $t \in \{0, \dots, \text{steps_per_day} - 1\}$ is the step-of-day index. The precomputed table is:

$$\text{per_step_reward}[t][s] = \text{reward}[s] \times \text{reward_multiplier_by_step}[t]$$

with a default multiplier of 1.0 for all t when not specified. Currently:

$$\begin{aligned} \text{reward}[\text{active}] &= 1.0 \\ \text{reward}[\text{sedentary}] &= 0.0 \end{aligned}$$

Single-timescale placeholder. Multi-timescale reward (immediate + delayed body measure) is deferred to Phase 2.

1.5 Episode Structure

$T = \text{episode_days} \times \text{steps_per_day}$ timesteps. Default: $T = 90 \text{ days} \times 5 \text{ steps/day} = 450$.

2 Agents

All agents are contextual bandits: state is accepted but not used in action selection. For the stationary MVP transition matrix, $E[r \mid a]$ is the relevant quantity.

2.1 Thompson Sampling

Beta-Bernoulli TS with prior $\text{Beta}(1, 1)$ per action. Posterior update:

$$\alpha_a \leftarrow \alpha_a + \mathbb{I}[s'.\text{activity} = \text{active}], \quad \beta_a \leftarrow \beta_a + \mathbb{I}[s'.\text{activity} = \text{sedentary}]$$

Action selected by sampling $\theta_a \sim \text{Beta}(\alpha_a, \beta_a)$ and choosing $a^* = \arg \max_a \theta_a$.

2.2 Epsilon-Greedy

$\epsilon = 0.1$. With probability ϵ , select uniformly at random; otherwise select $\arg \max_a Q(a)$. Q-values updated via incremental average:

$$Q(a) \leftarrow Q(a) + \frac{r_t - Q(a)}{n_a}$$

2.3 UCB1

Upper Confidence Bound with exploration parameter $c = 2.0$. Selects:

$$a^* = \arg \max_a \left[Q(a) + c \sqrt{\frac{\ln N}{N_a}} \right]$$

where N is total steps and N_a is the number of times action a was chosen. The confidence bonus shrinks as N_a grows, naturally shifting from exploration to exploitation. Each action is pulled once before UCB is applied.

2.4 Random Baseline

Uniform random action selection. No learning. Serves as lower bound.

3 Initial Results

All agents run over 10 random seeds (1–10), 450 timesteps each. Results are reported as mean $\pm$ standard deviation across seeds.

3.1 Standard Config (config/rule_based.yaml)

No `reward_multiplier_by_step` – uniform 1.0 multiplier on all steps.

Agent	Total Reward	Mean Reward/Step
Thompson Sampling ($\alpha_0 = \beta_0 = 1$)	160.3 ± 10.5	0.356 ± 0.023
Epsilon-Greedy ($\epsilon = 0.1$)	160.8 ± 15.3	0.357 ± 0.034
UCB ($c = 2.0$)	148.2 ± 15.0	0.329 ± 0.033
Random	135.6 ± 10.7	0.301 ± 0.024

Table 2: Results on `config/rule_based.yaml` (10 seeds, 450 timesteps). TS and ϵ -greedy perform similarly; TS has lower variance. UCB outperforms random but shows higher variance.

3.2 Masked Config (config/rule_based_with_mask.yaml)

`reward_multiplier_by_step`: `[1, 1, 1, 1, 0]` – step 4 (end of day) yields zero reward regardless of outcome. All agents drop proportionally to the masked fraction (1/5 of steps zeroed), confirming the multiplicative reward model.

Agent	Total Reward	Mean Reward/Step
Thompson Sampling ($\alpha_0 = \beta_0 = 1$)	126.6 ± 6.8	0.281 ± 0.015
Epsilon-Greedy ($\epsilon = 0.1$)	127.3 ± 10.2	0.283 ± 0.023
UCB ($c = 2.0$)	116.9 ± 8.6	0.260 ± 0.019
Random	107.8 ± 6.7	0.240 ± 0.015

Table 3: Results on `config/rule_based_with_mask.yaml` (10 seeds, 450 timesteps). Reward drops $\approx 20\%$ across all agents, matching the 1/5 masked fraction. Relative ordering is preserved.

4 Known Limitations

- Agents are state-blind bandits, not state-aware RL. State-aware agents are Phase 2.
- Single-timescale reward only. Multi-timescale (immediate + 3-week delayed) is Phase 2.
- No user response model. Burden accumulation and 4 archetypes are Phase 2.
- Transition matrix is fixed and known. Learned transitions from real data are Phase 2.

- Transition probabilities are placeholder values informed by Reactance theory. Calibration against real data is Phase 2.